

Activity 2 -Research Methodology

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Based on the identified phenomenon, carryout the following activities

- Identified phenomenon : Lightning

1. Describe the phenomenon using two main ontologies. Be critical about it

- **Ontology 1: Realism (Scientific Realism / Critical Realism)**

From a realist perspective, lightning exists as an independent physical reality, regardless of human perception. The phenomenon occurs due to natural processes, specifically, electrostatic discharges between clouds or between clouds and the ground. Its existence can be empirically verified using instruments such as lightning detectors, high-speed cameras, and electric field meters. This position argues that lightning is a universal truth governed by natural laws of physics, measurable and reproducible.

While realism provides a reliable scientific explanation, it tends to overlook cultural and contextual interpretations. By focusing only on objective measurements, it assumes one absolute truth and discounts the influence of human meaning-making.

- **Ontology 2: Relativism / Constructivism**

From a relativist or constructivist perspective, lightning is not only a physical event but also a socially and culturally constructed phenomenon. Different societies attach varied meanings to it—some interpret it as a divine sign, others as a symbol of power or danger. The “truth” about lightning, in this view, is shaped by language, culture, and human interaction. What one culture considers a scientific occurrence may, in

another, be considered a spiritual message.

While constructivism highlights human interpretation and meaning, it risks denying the underlying physical reality. If taken to extremes, it may disregard empirical evidence in favor of subjective or symbolic interpretations.

2. Establish epistemological definition for the chosen phenomenon

- **Objectivist Epistemology:**

Knowledge of lightning can be gained through empirical observation and measurement. Using instruments, scientists quantify lightning's electrical discharge, current strength, duration, and frequency. The goal is to establish objective, generalizable knowledge that holds true across contexts, independent of social interpretation. This aligns with positivist science, where knowledge is validated by reproducibility and empirical testing.

- **Constructivist Epistemology:**

Knowledge of lightning is also shaped by how people perceive, interpret, and communicate it within social contexts. Folklore, religious beliefs, and cultural narratives contribute to how communities “know” lightning. In this view, epistemology emphasizes subjective interpretation, where knowledge is co-created through experience and cultural context rather than discovered as a fixed external truth.

Epistemological Definition (Balanced View):

Lightning is a phenomenon that can be known both through objective measurement of its physical properties and through the interpretive frameworks provided by culture and society.

Knowledge is therefore dual in nature, scientifically verifiable on one hand, and socially constructed on the other.